

Japan is gripped by mass allergies. A 1950s project is to blame

A decision made 70 years ago to reforest vast swathes of Japan with just two kinds of tree has come back to haunt the country.

In February, videos showing what looked like waves of smoke blowing off an evergreen forest went viral in Japan. It wasn't smoke – it was pollen, and the videos were a warning to tens of millions of residents of the archipelago nation: prepare your masks and allergy medicine.

Every spring (which is already arriving earlier in Japan due to climate change) you'll see people of all ages wearing masks on the streets of cities across the country. The reason: hay fever, driven by all the pollen.

Hay fever – also known as allergic rhinitis – has now become a national crisis in Japan, with an estimated 43% of the population experiencing medium to severe symptoms. This compares to 26% in the UK and 12-18% in the US.

As well as the discomfort, these allergies can lead to sleep loss and poor concentration, and sufferers are more likely to experience other conditions such as asthma and food allergies. At the peak of Japan's hay fever season, the economic impact from both sick days and lower consumer spending is estimated at \$1.6bn (£1.2bn) per day.

So why does Japan have such bad allergies? The reason has little to do with poor health or pollution, or even the natural environment, but decisions made by leaders more than 70 years ago in the decades after World War Two.

During the war, oil and gas shortages led Japan to turn to the nation's most abundant natural resource – forests – as a source of fuel for home and industry. The result was widespread deforestation of natural forests, with the mountains around major cities like Tokyo, Osaka and Kobe completely stripped bare of trees.

"After World War Two, many of Japan's mountains became barren, causing disasters in various regions," says Noriko Sato, a professor and forestry researcher at Kyushu University in Fukuoka, Japan. (Bare mountains can increase the incidence of landslides and flooding). "Large-scale afforestation was carried out by public works, funded by tax revenues, to prevent soil erosion."

Aiming for rapid reforestation, the government chose to plant reams of only two different native, fast-growing evergreen species that could quickly reforest landscapes and provide

wood for future use in construction: the Japanese cedar, sugi, and the Japanese cypress, hinoki.

Today, these hinoki and sugi plantation forests still cover around 10 million hectares (25 million acres) – a fifth of Japan's entire land area.

The problem is, sugi and hinoki trees also produce large amounts of lightweight pollen which can easily drift into cities. It's this pollen, often released all at once from the monoculture plantations, that is responsible for most seasonal allergies in Japan. The issue has become all the worse since these trees release ever more pollen after maturing at 30 years of age – now the case for nearly all of them.

"Pollen allergies have become a national health issue in Japan," says Sato. "Addressing this problem is urgent."

In 2023, Japan declared allergies a national social problem and the central government set out an ambitious plan – reduce pollen by 50% in 30 years. As a first step, it aims to reduce the forest areas covered with high-pollen sugi trees by 20% by 2033.

But swapping out forests covering over 2% of Japan in 10 years is a massive endeavour. Plus, simply cutting these trees down won't be enough – they also need to be replaced with new forests to avoid soil erosion or accidentally undercutting Japan's own climate targets.

Walking through sugi or hinoki plantation forests is eerie – all the trees are the same height and there are few birds or insects. The ground is spongy with dry needles, and there's little light or sound.

It's a stark contrast to Japan's natural forests, which teem with biodiversity and sound. With their diverse tree species like red pine, larch and maple, these forests support more of all kinds of wildlife. Japan's unique geography has made it one of the world's biodiversity hotspots, but habitat loss and invasive species have led much of its unique wildlife to become increasingly at risk.

With the monoculture plantation forests causing so many problems, it makes sense that Japan is now trying to replace them with something better. But it's a daunting challenge. The reason? Japan has a lot of forests. In fact it's one of the most forested industrialised nations in the world, with forests covering 68% of its land, a third of which are sugi and hinoki plantations. The US, by contrast, is 34% forested; the UK just 13%.

Across Japan, forests can be found right by cities. Japanese even has a word for the transition area between city and forest: satoyama.

Still, even before the 2023 government declaration, some local actors and non-profits had begun efforts to turn these forests into biodiverse ecosystems, and some are already seeing the benefits. The small town of Nishiawakura, Okayama, for example, has created an entire economy around reducing the 84% of its forests made up only of hinoki and sugi, turning wood into heat for eel farms as well as chopsticks and timber.

In 2020, Kobe, a larger port city in central Japan with a dense urban core and vast forests within its city limits, began an effort to turn more than 180 hectares (445 acres) of plantation back into natural broadleaf forests in a 15-year cycle.

Every year, an area is selectively clear-cut, removing sugi, hinoki but also other invasive species like bamboo. Broadleaf trees are left, and with more sun coming through to the ground, they grow back, along with other new seedlings either planted by staff or brought by birds or animals.

With its scheme now about halfway done, local government workers say they have been pleasantly surprised by how quickly biodiversity has returned.

"Our wildlife monitoring is showing more animals and insects returning, including badgers, pond turtles, many species of frogs, and rare insects too, which is encouraging," says Atsushi Okada, head of the Kobe City Environmental Bureau.

As well as addressing the pollen issue, the scheme aims to fulfil the Kobe's pledge to increase its protected areas to 30% of all land by 2030. More diverse forests should also protect the city against the landslides and natural disasters poised to become more frequent due to climate change, says Daisuke Tochimoto, a forester with the City of Kobe.

As for the cut trees, they are used for heating, furniture production and Japanese white charcoal, a smoke-free barbecue fuel which could also be used in industrial processes. The hope is that, over time, the project can become financially sustainable and not reliant on public funds, says Okada.

Similar projects are beginning in other parts of Japan. One project in Hotani, Osaka, is now restoring wetlands and grasslands. And the largest effort aims to turn 10,000 hectares (25,000 acres) of plantation forests in Gunma prefecture to meadows and mixed deciduous woodland.

Smaller-scale projects are also common, says Akira Mori, a professor of biodiversity and ecosystem services at the University of Tokyo, pointing to dozens of initiatives around Japan.

Since the goal of removing 20% of the high-pollen sugi plantations was announced, the country has designated approximately 980,000 hectares (2.4 million acres) of sugi

plantation forests as areas for focused logging and replanting. Still, not all of this is being turned into broadleaf forests: some of it is fresh plantations, often planted with low-pollen or pollen free sugi.

Japan's ministry of agriculture, forestry and fisheries tells the BBC it plans to track the progress toward its targets for the reduction of high-pollen sugi-planted forests, but did not provide any numbers on progress so far.

Still, these efforts may not yet be large enough to make much of a difference to the pollen.

And even if it achieves the goal, 80% of the plantation forests will remain. So Japan is also trying other ways to tackle hay fever.

Pollen data and forecasts, for example, are being used to better understand where dispersion is likely, allowing authorities to selectively cut down the worst offending forests, and researchers are even looking at spraying trees with solutions to suppress pollen. In 2023, one forecasting company distributed thousands of pollen-detecting robots – whose eyes go different colours depending on pollen levels – across Japan.

Medicine is another prong to the attack, with the development of new treatments to better ease the symptoms of pollen exposure. One Japanese trial, for example, showed a long-acting under-the-tongue immunotherapy tablet was still helping alleviate symptoms two years after treatment. Other scientists have even been experimenting with genetically modified rice designed to alleviate allergy symptoms. (Read more about the new wave of effective cures for seasonal allergies).

When the sugi and hinoki forests were first planted in the 1950s and 60s, they weren't meant to stand forever. At the time, it was assumed they would be gradually cut down and replanted over time, as had been the case before the war. But as Japan's economy boomed in the late 60s and 70s, major cities like Kobe and Tokyo grew rapidly, and it ended up being cheaper to import wood from other countries such as Malaysia and Indonesia.

In 2011, though, Japan set a goal of relying less on forestry imports, and has seen its domestic wood use grow from 26% in 2010 to nearly 42% in 2020. Japan's forestry ministry says promoting the growing demand for sugi wood is another countermeasure it is taking against pollen sources, alongside improving its forestry productivity and workforce.

Of course, if Japan is going to exploit its forests, it has to avoid the same mistakes made in Southeast Asia, where cheap wood means the clear-cutting of tropical forests. Junichi Mishiba, forest project coordinator at the non-profit Friends of the Earth Japan, worries that more incentives to cut down trees is leading to bad environmental practices. "There is an increase in clear-cut areas resulting from policies promoting harvesting," he says.

To support its efforts to replace the plantations, in 2024 the national government began collecting a new forest environment tax of 1000 yen (\$6/£5) per year on all residents. The money is being used to support sustainable forestry, including reducing plantation forests and replacing older sugi with new, low-pollen seedlings, especially in urban areas.

Data on its impact is not yet available, but Mori argues the support is not enough, with municipalities often lacking the capacity and expertise to design and monitor such changes to forests. A 2023 report by the Forest Declaration Assessment noted that in recent years, only 30-40% of Japan's newly harvested land has been replanted.

Japan's forestry ministry says that it has since improved its methodology on how it calculates this harvesting area, and now calculates the figure to be approximately 50-60%.

Good forest management will be essential, agrees Mika Akesaka, an associate professor of economics at Kobe University. Leaving felled trees unmanaged, for example, can increase landslide risk and reduce water retention capacity, she says.

Mishiba, though, fears that by focusing only on seasonal allergies rather than wider ecological indicators, Japan is once again prioritising short-term solutions. The country needs to think 50 or even 100 years ahead, he says, considering biodiversity, climate and the role of the people who will live alongside these forests.

Japan's forestry ministry tells the BBC its forest planning system "requires that forest management be carried out with due consideration for the multiple functions of forests", therefore "no harm to the performance of [these forest] functions will be done by this policy initiative".

Urgency to act is also growing because of another unplanned factor – climate change. Around the world, temperature and weather shifts are impacting pollen spread and Japan saw its earliest pollen dispersal ever in 2025.

"Pollen dispersal is greatly influenced by weather conditions such as temperature and wind," says Mai Sato, a spokesperson with the Japan Weather Association (JWA), a forecasting company which releases regular pollen forecasts to the public.

Japan's vast forests also themselves hold huge amounts of carbon, and sugi plantations are responsible for almost half the carbon sequestered by its forests each year. Japan is leaning heavily on this carbon sequestration to achieve its net zero goal, and encourages it with a carbon credits scheme.

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Still, since 2004 Japan has seen a declining trend in the yearly amount being absorbed which it attributes to the maturity of its forests. Research has shown that since ageing trees absorb less carbon, thinning forests of old trees and planting new, younger and more diverse species will be essential to keeping Japan's forests an effective carbon sink.

Japan's forestry ministry says it is thoroughly replanting sites where sugi forests have been harvested with low-pollen cedar seedlings or other tree species. "Through this, we will continue to sustainably demonstrate the multiple functions of forests, including their role as a carbon sink, and contribute to Japan's climate targets," it says.

Before the 1960s, Japan didn't even have a word for hay fever. Japanese cedar pollinosis was first identified in 1963 and, according to researchers at the time, was new to the country. The hope is that with the return of more natural, diverse forests, Japan can one day go back to enjoying its springs – without the sneezes.

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*This article was updated with a response from Japan's ministry of agriculture, forestry and fisheries on 20/5/26.

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With storms occurring between 140 to 160 nights a year, it's no wonder the area is a world record holder.

In 1869, two companies connected the east to the west with the United States' first transcontinental railroad.

Under the surface of the ocean, 100,000 towering mountains dot the Earth. Their violent currents are key to life.

The ancient concrete made by the Romans is teaching modern construction some new tricks.

Princess Diana's cousin shares unseen photos of their private childhood.

This video has expired.

Eighty years ago, the US dropped a nuclear bomb on Japan, the only nuclear weapon ever used in warfare.

A woman who allegedly was the head of the Catholic Church became one of the most controversial Middle Ages tales.

Hiding in an attic, Jewish man Curt Bloch found inspiration through crafting anti-Nazi parody.

Watch how the maps and images of our planet from above have changed over the last two millennia.

From a shipwrecking yard in Bangladesh to a river of iron dioxide in Canada, a deep dive in Ed Burtynsky's work.

It starts nearly 40 years ago, when a teenage girl is pulled out of obscurity and thrust into the spotlight.

Snowfall is helping urban planners design safer and more pleasant streets in the US.

Valentine's Day is thought to celebrate romance but rude cards soured the holiday for its recipients.

An unexpected WW2 experiment by behaviourist B F Skinner proved that pigeons could be used for missile guidance.

In Sri Lanka, a charming elephant cheekily halts traffic for treats.

Watch two rhinoceroses involved in a game of 'kiss and chase'.

Dr. Lisa Dahlgren discusses the effect human touch has on our health and wellbeing.

While her three offspring take a leisurely bath, this Bengal tiger mother must find food for the entire family.

Watch red foxes challenge the Steller's sea eagle, the world's heaviest raptor, as they search for food in Japan.

The events will include bat nights, fungi and birdsong events and a summer solstice celebration.

A new DNA test could help screen whether patients require the treatment or not, according to a new international trial.

A woman is taking part in Walk for Wards, a fundraising event for the Salisbury Hospital charity.

The head was hung on the wall at the White Horse Inn at Susworth near Scunthorpe.

If you're having ice cream headaches often, it might be worth seeing if it runs in your family and taking a second look at your non-brain-freeze headaches too.

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US, UK and Australia to develop underwater drone technology

The US, UK and Australia say they will develop underwater drone technology to protect undersea cables and boost defence, under their military alliance known as Aukus.

The uncrewed undersea vehicle (UUV) technology is expected to be ready by next year. While the project's total cost was not stated, UK Defence Secretary John Healey said his nation would contribute £150m (\$201m).

The announcement, made by the countries' defence ministers at a security summit in Singapore, follows claims of slow progress in Aukus's projects.

Acknowledging the criticism, Healey said "for too long in Aukus, we talked too much and delivered too little", adding "that has now changed under our three governments".

The Aukus defence pact, which began in 2021, affirmed the three countries would develop nuclear submarines and share military expertise.

It is widely seen as a way to counter China's growing maritime presence in the Indo-Pacific and its role in rising tensions in disputed territories such as the South China Sea.

The UUV technology is the first signature project under Aukus's Pillar Two, in which the partner countries work together on "advanced capabilities" in areas such as long-range hypersonic missiles, undersea robotics and AI.

A joint statement said the new project would see "cutting edge payloads and enabling systems" developed for UUVs that could protect seabed infrastructure, conduct strikes, and conduct surveillance, reconnaissance and logistics operations.

Healey also said that sensors and weapons systems would be developed for the UUVs, which would "rapidly give our forces advanced battle technologies".

It would also help them deal with threats "including to our underwater cables and pipelines on which so much of our daily life depends". Such efforts would strengthen deterrence in the Pacific, Atlantic and waters in the High North, he added.

The announcement comes a month after Healey accused Russia of running a covert operation over cables and pipelines in waters north of the UK. Moscow has denied the allegations.

In December, the UK and Norway signed a pact to hunt Russian submarines in the North Atlantic to protect undersea cables.

The UK is connected by about 60 undersea cables, which British officials say are increasingly under threat from Moscow, with a 30% rise in Russian vessels spotted in UK waters over the past few years.

Elsewhere, Chinese ships are suspected to have damaged undersea cables in waters surrounding Taiwan and in Swedish territory.

There have also been multiple reports of cables damaged in the Baltic Sea.

On Saturday, the three defence ministers did not respond to a question from the BBC on whether the UUV technology project was aimed at countering Russian and Chinese undersea activities.

They also did not answer a question on whether progress on Aukus projects was too slow.

Under Pillar One of the defence pact, nuclear-powered attack submarines would be built in the UK and Australia for use in their navies.

For Australia in particular, the deal represents a major upgrade to its military capabilities.

The country will become just the second to receive Washington's elite nuclear propulsion technology after the UK, which began receiving it decades ago.

But questions have been increasingly asked in Australia about whether the country's biggest-ever defence project could be achieved in time to replace their ageing submarines - or if at all.

The Aukus submarines are only scheduled to be ready in the 2040s. In the meantime, the US and UK will be rotating their existing nuclear-powered submarines through Australia, and in the 2030s Australia will buy second-hand nuclear submarines from the US.

Days before arriving in Singapore for the Shangri-la Dialogue, Australia's Defence Minister Richard Marles addressed this criticism, saying they had to go ahead with the Aukus submarine project as there was no "plan B".

On Saturday, US Defence Secretary Pete Hegseth said the plan to rotate US and UK nuclear-powered submarines through Australia was "still on track", with the first US Navy personnel due to arrive later this year.

Marles said that the HMAS Stirling navy base in Western Australia would be ready to host the rotational submarine force by the end of 2027, and that "work is at a pace" to establish a construction yard in South Australia that would build the Aukus submarines.

Performers Young MC, Poison frontman Bret Michaels and country singer Martina McBride raised concerns the event will be political.

Every president in modern history - including Donald Trump this week - has taken an annual physical exam that is as much about messaging as it is about health.

The US president met his advisers, after officials confirmed the US and Iran had agreed a framework of a deal.

The White House releases the results of the US president's latest medical examination, which states he is "fully fit" to carry out his duties.

The testimony comes about a month after America's top prosecutor was ousted by US President Donald Trump.

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'No-one feels safe now': Residents of Romanian city hit by drone share fears

In some parts of Europe, Russia's full-scale invasion of Ukraine can feel like a distant threat. But in Romania, that war is right next door and increasingly dangerous.

In Galati, there is an apartment block with a hole in the roof that proves it.

Residents have just begun returning to check on their homes, after an attack drone slammed into the building early on Friday as dozens of people slept.

It sparked a fire and panic.

We climbed 11 floors up to the roof on Saturday to see where the drone punched through the concrete. There's a jagged hole, a couple of metres wide, now covered with plastic.

The flat below was badly damaged, and a woman and her teenage son remain in hospital with bruises and minor burns.

But it's clear the consequences of this strike could have been far worse: the drone hit the lift shaft on the roof, which absorbed much of the blast.

"It was really very terrifying," says Costel Patrichi, a resident who's in charge of the building. "But if the drone had hit the side, it could have destroyed a whole floor or more."

He describes how his phone buzzed with an alert that morning just before 02:00, warning of the danger: a drone was approaching from the Ukrainian border a few miles away.

Moments later came the bang.

"They told us we are protected by Nato, not to worry. But look where we are now!" Costel tells me, frustrated like many that Romania's air force couldn't intercept the drone.

When a Ukrainian drone targeting northern Russia was recently knocked off course into Estonia, it was a Romanian fighter jet there that shot it down - part of Nato's quick reaction force.

Here, though, pilots only had moments to react before the weapon was over a built-up area. At that point, interception was too risky.

"Now I'm afraid. If go back to my flat tonight, I will sleep with fear. Because this could happen again," Costel admits.

It is the same fear that Ukrainians endure nightly as Russia launches ever more attack drones at its neighbour. Very often, they smash into residential areas, destroying homes and taking lives.

Now Romania, a member of both Nato and the EU, has been hit.

It is the most serious incident of its kind in this country since Russia's full-scale invasion began in 2022.

True to form, Russian President Vladimir Putin claims there is no evidence this was a Russian drone.

But Romania has been very clear: it was a Geran-2, otherwise called a Shahed, and it was Russian.

"It's sure, because we had another one four or five weeks ago that didn't explode. We compared and they are completely identical," Romania's President Nicosur Dan told the BBC World Service.

The drones are used to target Ukrainian ports on the other side of the river Danube that are vitally important to Ukraine's grain exports.

On Friday, Romania tracked a swarm of 43 of them as they travelled from east to west.

"One hit by the Ukrainian army changed direction and passed to Romanian territory. That is sure," Dan said.

Romania's Nato allies have called Russia's conduct "reckless" and stressed that Moscow's war of aggression was to blame for what happened.

In Washington, though, US Secretary of State Marco Rubio ignored reporters' requests to comment.

And there is clearly caution in the response, as well as condemnation.

No-one is accusing Moscow of mounting a deliberate attack on Romania.

And whilst government sources in Bucharest tell us they considered invoking Article 4 of the Nato treaty, which would trigger an emergency meeting, that idea was rejected to avoid creating panic.

The next potential step would have been Article 5: the mutual defence clause, under which an attack on one member is considered an attack on all.

That's not on the table.

Instead, Romania has shut down a Russian consulate in the port city of Constanta as a "warning", according to its president. Dan said the next move in the "diplomatic hierarchy of measures" would be to kick out the Russian ambassador.

But for now, he's going nowhere.

Romania has called for Nato to move faster with a pledge to transfer more military equipment to this stretch of its eastern edge.

The government is already acquiring drones of its own and has plans to develop others in co-operation with Ukrainian companies.

The EU was already working on a new set of sanctions against Moscow.

But the risk of this war escalating and expanding has rarely felt greater - and the people we met in Galati feel very vulnerable.

"This was insane, it happened right in the middle of town," says Adrian, after checking his own family's flat in the building that was hit.

"No-one feels safe now."

For that, he blames Russia and its president.

"But I don't think the sanctions are enough," Adrian adds. "Because they could take everything from Russia, and they would still attack."

Additional reporting by Mircea Barbu

From enigmatic KGB agent to wartime ruler, this is how Putin has repeatedly reinvented his image, and himself.

BBC Verify has analysed videos of attacks in occupied Ukraine on Russian trucks carrying ammunition, fuel and food.

The US is considering options for punishing Nato allies which it considers to have failed to offer support during the Iran war, according to a Pentagon memo seen by Reuters.

With the US pulling out of trilateral talks with Russia and Ukraine, the EU is looking for potential candidates to step in.

Moscow's rhetoric may point to the Kremlin's nervousness over the war's direction, our defence correspondent writes from Kyiv.

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Oscar-winning Star Wars editor Marcia Lucas dies aged 80

Marcia Lucas, the Oscar-winning editor of the original Star Wars film, has died aged 80.

Lucas, who was married to Star Wars creator George Lucas during the making of the first three films, was regarded as a pivotal creative force behind the space saga's early success, imbuing the original series with emotional depth and narrative clarity.

She died from metastatic cancer at her home in Rancho Mirage, California, on Wednesday surrounded by loved ones, according to her family.

"Marcia was a force," her family said in a statement to US media on Friday. "A true trailblazer for women in film and one of the most influential editors in cinematic history; she helped redefine what film editing could be."

Lucas won the Academy Award for Best Film Editing for 1977's Star Wars - later renamed A New Hope - alongside editors Richard Chew and Paul Hirsch.

Although her contributions largely took place behind the scenes, her role in shaping the film's emotional heart and narrative structure has been widely recognised in the decades since its release.

George Lucas credited her with helping make sense of the vast amount of footage filmed for the climactic Death Star battle sequence.

"It was extremely complex and we had 40,000 feet of dialogue footage of pilots saying this and that," he told Rolling Stone shortly after the film's release.

"Nobody really has ever tried to interweave an actual plot story into a dogfight, and we were trying to do that."

Born Marcia Griffin in Modesto, California, in 1945, she began her career as a film librarian before becoming one of Hollywood's most respected editors.

After marrying George Lucas in 1969, she worked on several of his early films, including THX 1138 and American Graffiti - earning an Oscar nomination for the latter.

She also collaborated with director Martin Scorsese on a string of his acclaimed 1970s films including Alice Doesn't Live Here Anymore, Taxi Driver and New York, New York.

Lucas later returned to the Star Wars franchise, working on The Empire Strikes Back in 1980 and Return of the Jedi in 1983.

She and George Lucas adopted a daughter, Amanda, in 1981. The couple divorced in 1983 after 14 years of marriage.

She later married Tom Rodrigues, a production manager at Skywalker Ranch, with whom she had a second daughter, Amy.

Her family said in its statement: "Her influence on film is indelible, but those who knew her best will remember the way she made life feel more vivid, more beautiful, more fun and more full of love.

"Her work was known for its emotional intelligence, rhythm and humanity - a rare ability to find the truth of a scene and bring heart, momentum and clarity to the screen."

"I love film editing," Lucas once told a reporter, according to Lucasfilm.

"I have an innate ability to take good material and make it better, and to take bad material and make it fair."

Paying tribute on Saturday, Lucasfilm said it was "deeply saddened" to learn of her death, adding it "joins the global filmmaking community in mourning the loss of Marcia Lucas".

Meanwhile, Mark Hamill, who portrayed Star Wars protagonist Luke Skywalker, wrote that he and his wife Marilou were "deeply saddened by the loss of our lifelong friend".

He added: "Not just a gifted, innovative artist, she also happened to be a genuinely nice person. Smart, funny and just plain fun to be around. Thankfully, her memory lives on and we will never stop missing her."

The Welsh screenwriter says intimacy co-ordinators offer protection to actors with less power.

The Amazon Prime adaptation of Elle Kennedy's ice hockey romance book is an early hit with viewers.

YouTuber Kane Parsons' eerie YouTube series has been adapted for a film called Backrooms.

Fresh revelations come after two women told the BBC they had been raped on the Channel 4 reality show.

The cobbled streets of Hull are standing in for the East End of London for a new film set in 1888.

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How an enslaved, shipwrecked African became the US's first great explorer

Nearly 500 years ago, a Moroccan man walked thousands of miles from Florida to the Pacific Coast, becoming the first known outsider to see the American West.

In 1528, a man from Morocco washed up on the coast of present-day Texas, more dead than alive. He had spent the previous month adrift in the Gulf of Mexico alongside a group of Spanish sailors on a flimsy lifeboat lashed together with tree trunks, horse hide and what was left of their tattered clothes. When a storm stranded the castaways on a barrier island near Galveston, they unwittingly became the first people from the Old World to enter the American West – and when they did, they were each starving, exhausted and naked.

In the weeks that followed, the shipwrecked survivors began dying, one by one. Many succumbed to hunger, others to the elements and some to attacks from Indigenous tribes. Of the roughly 600 men who had set sail from Spain a year earlier on this ill-fated expedition to conquer present-day Florida and the Gulf Coast for the Spanish Crown, only four survived: three Spanish captains and, somehow, the enslaved Moroccan.

During the next eight years, the man would become the party's de facto leader, and embark on one of the most remarkable survival journeys in exploration history. And yet, we don't even know his real name.

Known variously as Esteban de Dorantes, Esteban the Moor or – most commonly – Estevanico, this enigmatic individual was one of the first documented Africans, Arabic speakers and Muslims to step foot in what is now the United States, arriving nearly 40 years before the first European settlement. Between 1528 and 1536, he walked roughly 2,250 miles (3,620km) west from Florida to the Pacific Coast of Mexico, completing what is widely believed to be the first recorded crossing of North America in history and predating Lewis and Clark's overland expedition to the Oregon Coast by nearly 300 years.

Along the way, Estevanico was captured by Native Americans, learned their languages and became a healer before journeying an additional 1,300 miles (2,090km) south with the three other shipwreck survivors from the Gulf of California to Mexico City. He then embarked on a separate 1,500-mile (2,415km) odyssey north, and became the first known non-Native American to enter modern-day New Mexico and Arizona.

"Estevanico is one of the most extraordinary, yet overlooked, figures in the early history of what would become the American Southwest," said Dr Hsain Ilahiane, an anthropologist and professor at the University of Arizona, who has spent years studying the explorer. "He

helped open routes, trails and geographic knowledge that later informed Spanish incursions into [the present-day American West]."

Still, most people have never heard of him – even in the US.

Now, as the US celebrates its 250th anniversary and looks back on its origins, a growing number of museums, tours and monuments around the country are now highlighting his little-known legacy.

Since Estevanico left no written records, historians have pieced together his life largely through the surviving accounts of the Spanish survivors who travelled beside him. He was born in the early 1500s in Azemmour and enslaved by Spanish nobleman Andrés Dorantes de Carranza, who brought him on Spain's Narváez expedition to the Americas. Since Muslims were forbidden from travelling to the New World on official Spanish expeditions, Dorantes baptised him, renaming him Estevanico.

The 600-person, five-ship voyage set sail in June 1527, and it was a disaster from the start. Some 140 men deserted the expedition during a stop in Santo Domingo, and while resupplying in Cuba, a hurricane sank two ships and killed 50 more sailors. The crew eventually tried to sail to Mexico, but storms blew them into modern-day St Petersburg, Florida, in April 1528.

The expedition's leader, Panfilo de Narváez, then ordered Estevanico and several hundred men to march north to explore Florida's interior. After slogging some 300 miles (480km) through mosquito-infested swamps to what is now Saint Marks, Florida, and getting ambushed by Apalachee Native Americans, Estevanico and the remaining Spaniards were so decimated and desperate that they slaughtered and ate their last horses, built five makeshift rafts and sailed along the coast in hopes of reaching present-day Mexico.

As the men drifted west, they became the first non-Indigenous travellers to see the mouth of the Mississippi River. But the same storms that eventually capsized their rafts near Galveston would kill two-thirds of the remaining Spaniards, including Narváez himself.

American origins

This is the first in a four-part series highlighting the many ways the US was founded as it celebrates its 250th anniversary.

Like many of the shipwrecked castaways, Estevanico was soon captured by Native Americans. Somehow, the person with the lowest status in the Old World managed to survive in the New World.

"When the order of the Old World fell away, Estevanico had some advantages. Unlike [the Spanish], he had to speak other languages," said Laila Lalami, whose novel, *The Moor's Account*, is based on Estevanico and was a finalist for the Pulitzer Prize. "Since Azemmour was an Amazigh [city] controlled by the Portuguese, we know that he likely spoke Tamazight, Arabic and Portuguese. And since he was enslaved to a Spaniard, he spoke Spanish."

During his roughly five years of hard labour under Karankawa Natives, Estevanico learned their spoken language and the sign language shared by other groups in the region. He then covertly coordinated with a rival tribe of the Karankawa to escape with Dorantes, and helped free the two remaining Spaniards.

In the next two years, the four survivors walked from the Gulf Coast to the Gulf of California, relying on Estevanico's linguistic skills and cultural adaptability to navigate them from one Native community to the next.

"He was the leader of the group – the translator, the mediator, the scout," Ilahiane said. "They were moving in a new world and didn't know where they were going and Estevanico had an advantage: information."

During their travels, Estevanico met an Indigenous trader and arranged for them to stay in his community. While there, a sick person approached the visitors and asked them to treat her. By blending Christian rituals from the Spanish and Native practices Estevanico had observed, the men somehow convinced the woman that she was cured.

As word of the men's alleged healing powers spread, Indigenous communities began seeking them out and even following them as they journeyed west. Estevanico travelled ahead of the Spaniards to announce their arrival, often adorned with seashell bracelets on his arms, jingling bells around his ankles and carrying a rattle made from a dried gourd.

"He reminds me of a rural Moroccan Sufi with medicinal knowledge travelling from village to village, but he also picks up these [cultural] aspects from Native Americans and uses them as an entry point into other tribes as he wanders," Ilahiane said. "In the end, that would be a bit tragic for him."

After conquering the Aztecs in 1521, Spain had established a colony in modern-day central and southern Mexico. As the four travellers approached Culiacán in what is now Sinaloa, they met a group of Spanish horsemen who led them 1,300 miles (2,090km) south to the capital of New Spain (today, Mexico City). There, the four shipwreck survivors told tales of gilded cities somewhere in the northern desert that Native tribes had recounted to Estevanico – likely to get the outsiders to keep moving away from their villages.

Where travellers can learn about Estevanico

- The National Museum of African American History and Culture (Washington DC)
- National Museum of the American Indian (Washington DC)
- Arab American National Museum (Dearborn, Michigan)
- The Hidden History Museum (Los Angeles, California)
- Discover Florida Tours (St Petersburg, Florida)
- El Paso Museum of History (El Paso, Texas)
- New Mexico History Museum (Santa Fe, New Mexico)
- African American Museum of Southern Arizona (Tucson, Arizona)
- Coronado National Memorial (Hereford, Arizona)

Determined to locate these so-called "Seven Cities of Gold", the Viceroy of New Spain dispatched Estevanico to guide a group of Spanish friars north into modern-day New Mexico and Arizona in 1539. With turquoise jewellery around his neck and feathers in his hair, Estevanico walked ahead of the party, serving as the lead scout to gain knowledge of these fabled cities from Native tribes. If what he learned was moderately important, he'd send a small cross back to the Spaniards via a messenger. If it was important, he'd send back a medium-sized cross. And if it was highly important, he'd send back a large cross.

One day, shortly after becoming the first known outsider from the Old World to enter the land of the Zuni Pueblo people, Estevanico reportedly sent back a cross said to be "the height of a man". Elated, the Spaniards raced ahead to join him, only to learn that when he'd attempted to enter the southernmost of the supposed Seven Cities of Gold, Hawikuh, he had been killed by Zuni Natives.

While the Spanish never found any gilded cities, Estevanico's final journey led them into what became known as the "Tierra Nueva" (New Land), that would later become central to Spanish expansion in the American Southwest.

"Estevanico helped shape the geographic imagination of the Spanish Empire in North America," said Ilahiane. "His travels spread knowledge about routes, peoples and possibilities for expansion into regions that would become Texas, New Mexico, Arizona and beyond. This final journey into Zuni territory in 1539 triggered the Coronado Expedition of 1540 [that paved the way for Spanish expeditions and settlements across the American Southwest]."

From introducing devastating diseases and displacing Native populations to establishing Catholicism, cultural and culinary practices, this Spanish expansion would profoundly shape the future of the nation to come.

For centuries, Estevanico remained little more than a footnote in Spanish chronicles, and a forgotten figure among Americans.

"That's because his exploration isn't tied to the foundational myth of the United States," said Lalami. "It's one of those things that doesn't quite fit neatly into the history of the US's founding that's taught in schools about the 13 colonies and people escaping persecution in England."

But in the past decade, museums, monuments and tours around the US have increasingly begun recognising his role in the continent's early history.

"He had one of the most amazing journeys into the unknown in history. Someone should really make a movie about him already," said David Anderson, the owner of Discover Florida Tours, which has led historical guided walks in St Petersburg where Estevanico landed since 2017 and recounts his epic odyssey.

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In 2016, a 7ft (2.15m) bronze statue of Estevanico was erected at the Texas State Capitol in Austin, commemorating his role as the first African to step foot in Texas. Since 2022, the El Paso Museum of History in West Texas has recounted Estevanico's odyssey on its interactive digital wall, and the explorer is heavily featured throughout the New Mexico History Museum in Santa Fe and at the Coronado National Memorial in Arizona.

"This is someone who did something quite formidable," said Diana Abouali, the director of the Arab American National Museum in Dearborn, Michigan. "At the museum, we present him at the start of our 'Coming to America' exhibit to show that Arabic speakers are not new to this country. Arabic language and culture has been in what is now the United States for a long, long time."

For Lalami, what makes Estevanico so important isn't just what he did – it's what he represents.

"He was the first African to cross North America, but what makes him so fascinating is his position: he didn't come to the US as a conqueror, and he wasn't part of the people who were conquered. He was something in between," she said.

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